

CARVIDON-MR (TRIMETAZIDINE DIHYDROCHLORIDE MODIFIED RELEASE TABLETS 35 mg)

NAME AND STRENGTH OF ACTIVE INGREDIENT:

Trimetazidine Dihydrochloride 35 mg.

PRODUCT DESCRIPTION:

Light pink colored, circular, biconvex, film coated tablets, plain on both the faces.

PHARMACOLOGY:

Pharmacodynamics:

Mechanism of action

Trimetazidine inhibits β -oxidation of fatty acids by blocking long-chain 3-ketoacyl-CoA thiolase, which enhances glucose oxidation. In an ischemic cell, energy obtained during glucose oxidation requires less oxygen consumption than in the β -oxidation process. Potentiation of glucose oxidation optimizes cellular energy processes, thereby maintaining proper energy metabolism during ischemia.

Pharmacodynamic effects

In patients with ischemic heart disease, Trimetazidine acts as a metabolic agent, preserving the myocardial high-energy phosphate intracellular levels. Ant ischemic effects are achieved without concomitant hemodynamic effects.

Pharmacokinetics:

Orally, the maximum concentration is on average five hours after taking the tablet. About 24 hours, the plasma concentration is maintained at concentrations greater than or equal to 75% of the maximum concentration for 11 hours.

The steady state is reached, no later than the 60th hour.

The pharmacokinetic characteristics of trimetazidine are not influenced by meal.

The apparent volume of distribution is 4.8 l/Kg, the determination of trimetazidine protein is weak: the value measured in vitro is 16%.

The elimination of trimetazidine is mainly via the kidneys, mainly in the form of the unchanged product.

The half-life of trimetazidine is an average of 7 hours in healthy young volunteers, and 12 hours in the elderly over 65 years.

The total clearance of trimetazidine is the one that results from renal clearance majority directly related to creatinine clearance and to a lesser value, a hepatic clearance that decreases with age.

A specific clinical study, conducted in an elderly population, with a dose of 2 tablets per day in two doses, as measured by population kinetic data showed an increase in plasma exposure did not justify any change in dosage.

INDICATIONS:

Trimetazidine is indicated in adults as add-on therapy for the symptomatic treatment of patients with stable angina pectoris who are inadequately controlled by or intolerant to first-line antianginals therapies.

RECOMMENDED DOSE:

Posology

The dose is one tablet of 35mg of Trimetazidine twice daily during meals.

The benefit of the treatment should be assessed after three months and Trimetazidine should be discontinued if there is no treatment response.

Special populations

Patients with renal impairment

In patients with moderate renal impairment (creatinine clearance [30-60] ml/min), the recommended dose is 1 tablet of 35mg in the morning during breakfast.

Elderly patients

Elderly patients may have increased Trimetazidine exposure due to age-related decrease in renal function. In patients with moderate renal impairment (creatinine clearance [30-60] ml/min), the recommended dose is 1 tablet of 35mg in the morning during breakfast.

Dose titration in elderly patients should be exercised with caution.

Paediatric population:

The safety and efficacy of Trimetazidine in children aged below 18 years have not been established. No data are available.

Route of administration:

Oral.

CONTRAINDICATIONS:

Hypersensitivity to the active substance or to any of the excipients.

- Parkinson disease, Parkinsonian symptoms, tremors, restless leg syndrome, and other related movement disorders,
- Severe renal impairment (creatinine clearance < 30ml/min).

WARNING AND PRECAUTIONS:

Trimetazidine can cause or worsen parkinsonian symptoms (tremor, akinesia, hypertonía), which should be regularly investigated, especially in elderly patients. In doubtful cases, patients should be referred to a neurologist for appropriate investigations.

The occurrence of movement disorders such as parkinsonian symptoms, restless leg syndrome, tremors, gait instability should lead to definitive withdrawal of trimetazidine.

These cases have a low incidence and are usually reversible after treatment discontinuation. The majority of the patients recovered within 4 months after trimetazidine withdrawal. If parkinsonian symptoms persist more than 4 months after drug discontinuation, a neurologist opinion should be sought.

Falls may occur, related to gait instability or hypertension, in particular in patients taking antihypertensive treatment.

Caution should be exercised when prescribing trimetazidine to patients in whom an increased exposure is expected.

Moderate renal impairment

Elderly patients older than 75 years old

INTERACTIONS WITH OTHER MEDICAMENTS: No drug interactions have been identified.

PREGNANCY & LACTATION:*Pregnancy*

Studies in animals have not demonstrated a teratogenic effect; however, in the absence of clinical data, the risk of malformation cannot be excluded. Therefore, for safety reasons, prescription should be avoided during pregnancy.

Lactation

In the absence of data on excretion in breast milk, breastfeeding is not recommended during treatment.

ADVERSE EFFECTS:

Adverse reactions, defined as adverse events considered at least possibly related to Trimetazidine treatment are listed below using the following convention frequency: very common ($\geq 1/10$); common ($\geq 1/100$ to $< 1/10$); uncommon ($\geq 1/1,000$ to $< 1/100$); rare ($\geq 1/10,000$ to $< 1/1,000$); very rare ($< 1/10,000$); not known (cannot be estimated from the available data).

System Organ Class	Frequency	Preferred Term
Nervous system disorders	Common	Dizziness, headache
	Not known	Parkinsonian symptoms (tremor, akinesia, hypertonia), gait instability, restless, leg syndrome, other related movement disorders, usually reversible after treatment discontinuation
	Not known	Sleep disorders (insomnia, drowsiness)
Cardiac disorders	Rare	Palpitations, extra systoles, tachycardia
Vascular disorders	Rare	Arterial Hypotension, Orthostatic hypotension that may be associated with malaise, dizziness or fall, in particular in patients taking antihypertensive treatment, flushing
Gastrointestinal disorders	Common	Abdominal pain, diarrhoea, dyspepsia, nausea and vomiting
	Not known	Constipation
Skin and subcutaneous tissue disorders	Common	Rash, pruritus, urticaria.
	Not known	Acute generalized exanthemous pustulosis (AGEP), angioedema
General disorders and administration conditions	Common	Asthenia
Blood and lymphatic system disorders	Not known	Agranulocytosis Thrombocytopenia Thrombocytopenic purpura
Hepatobiliary disorders	Not known	Hepatitis

OVERDOSAGE & TREATMENT:

Very limited information is available on Trimetazidine overdose. Treatment should be symptomatic.

STORAGE CONDITION:

Store below 30°C. Keep away from the reach of children.

SHELF LIFE:

36 months from the date of manufacture

DOSAGE FORMS AND PACKAGING AVAILABLE:

Dosage form: Tablets

Packaging: Blister Pack of 2 x 10 Tablets (20 Tablets), 6 x 10 Tablets (60 Tablets) and 10 x 10 Tablets (100 Tablets).

Manufactured by:

MICRO LABS LIMITED,
92, SIPCOT, Hosur – 635 126,
Tamil Nadu, India.

Date of Revision of Package Insert: January 2016.